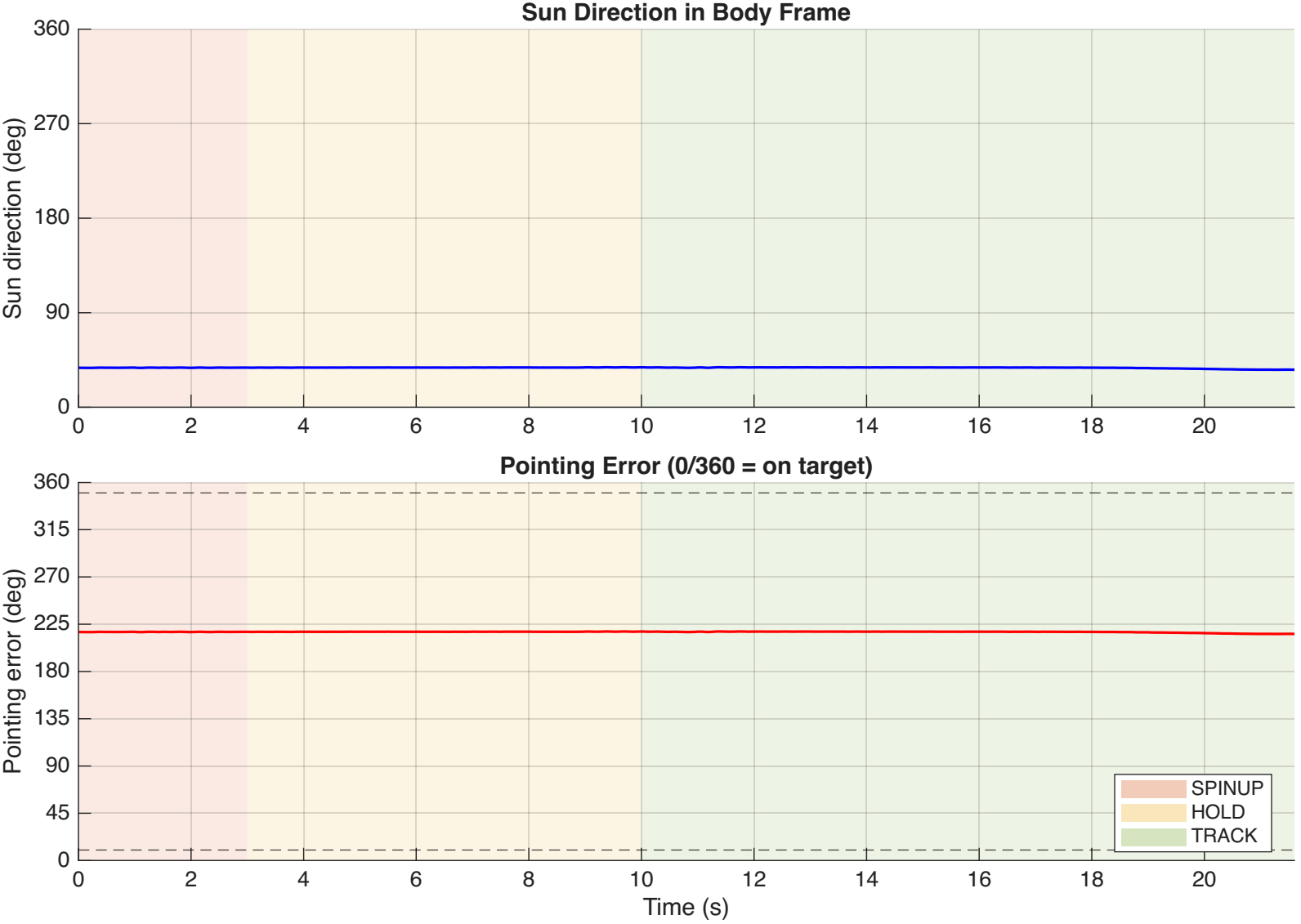
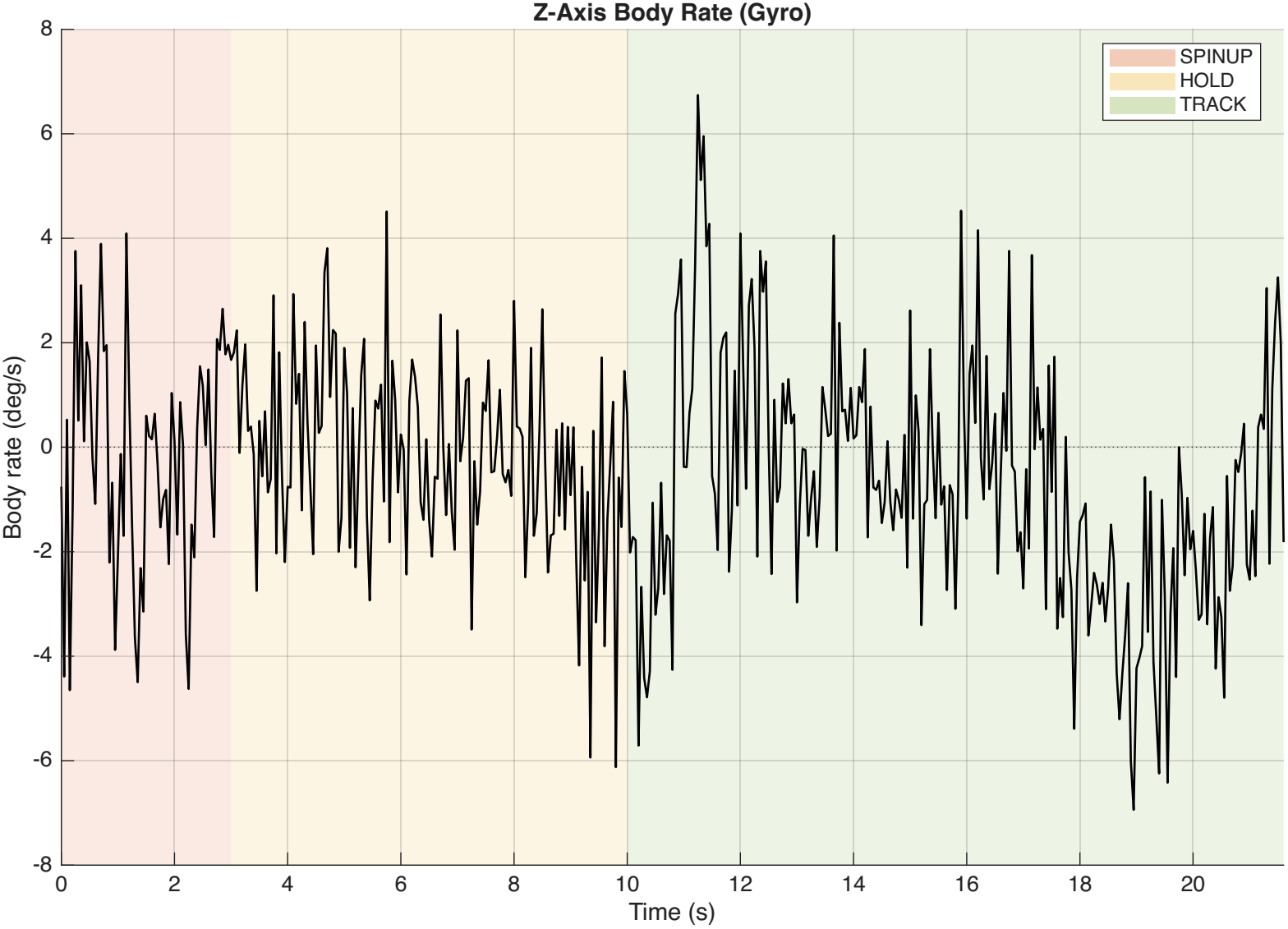
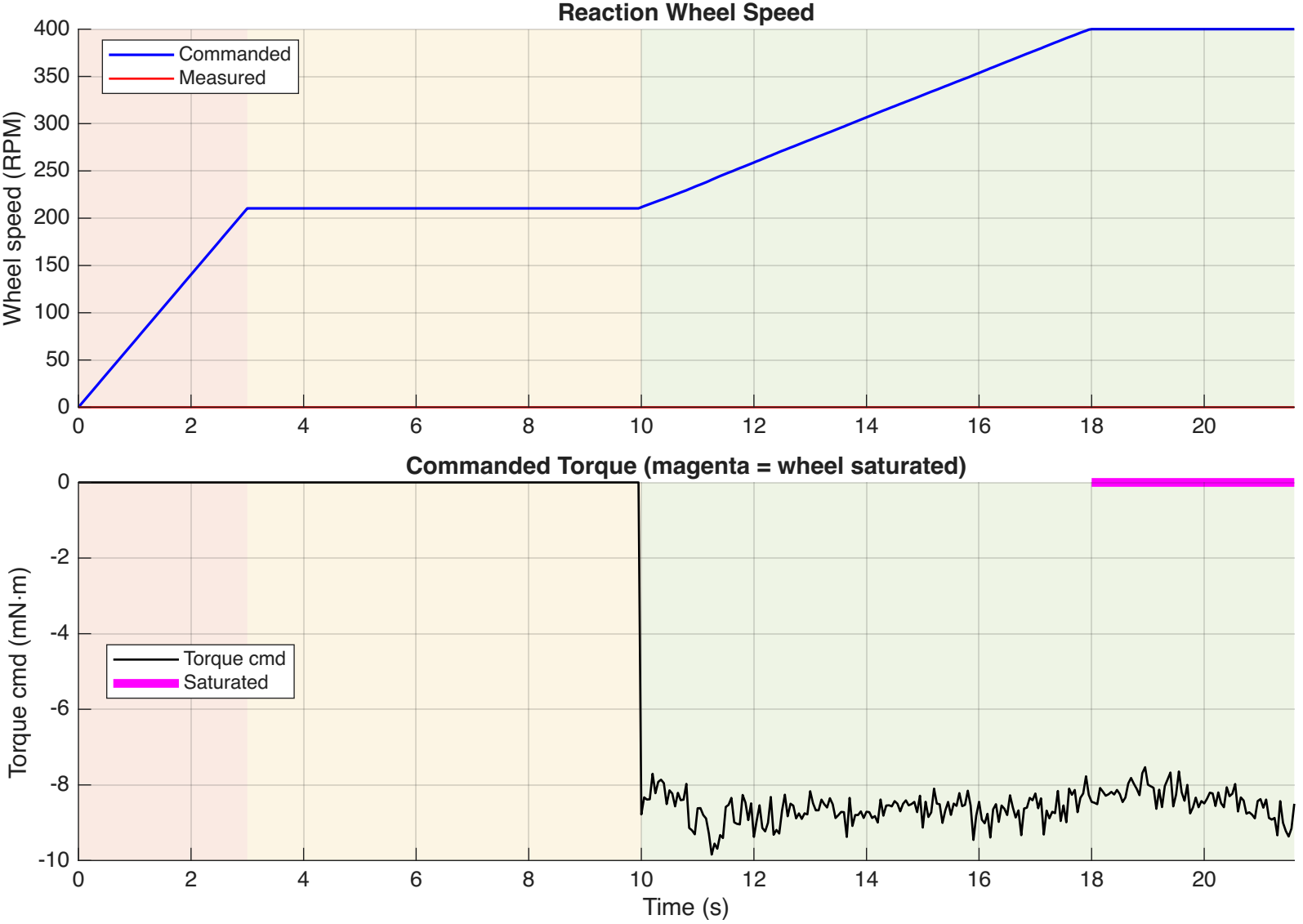
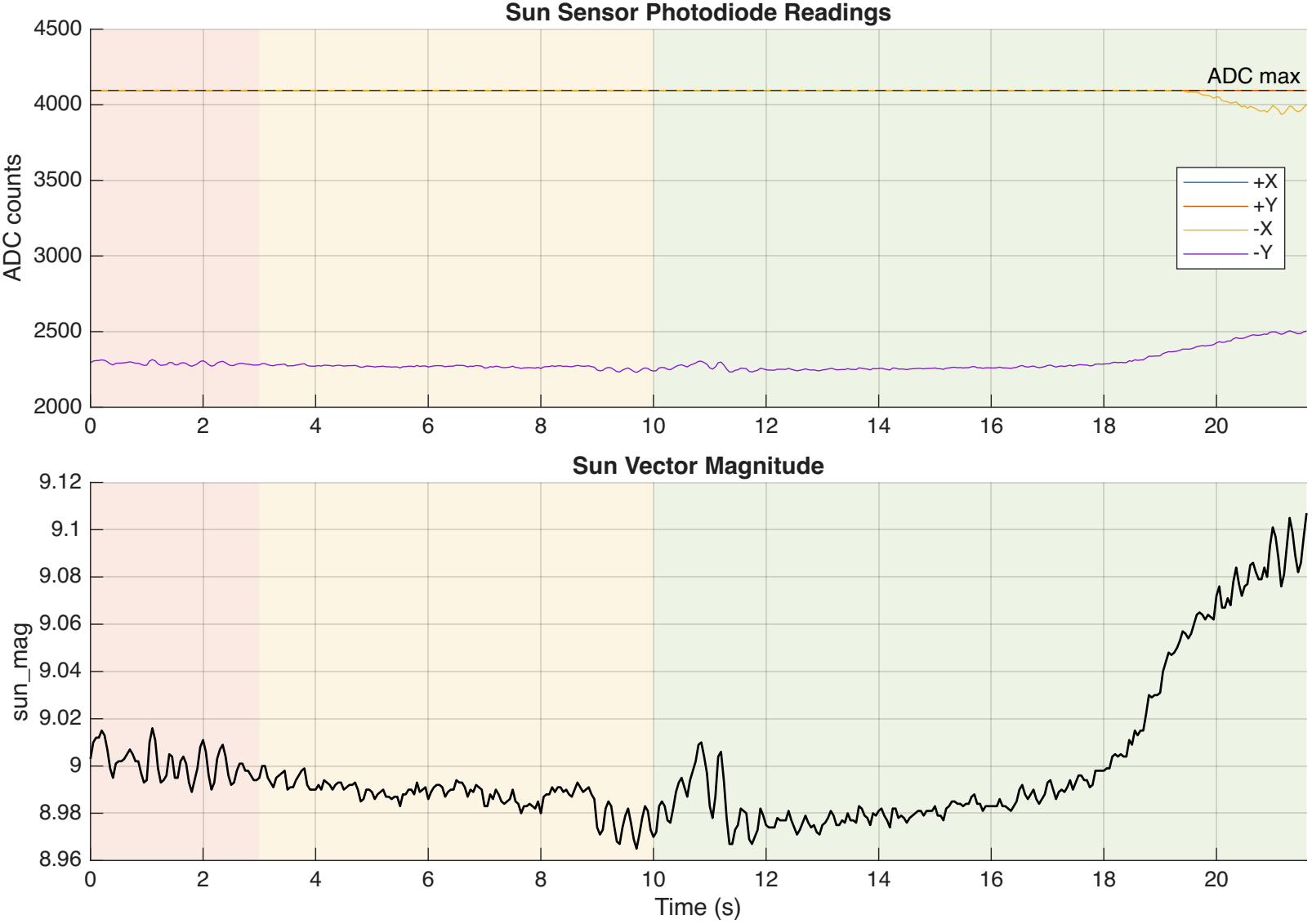
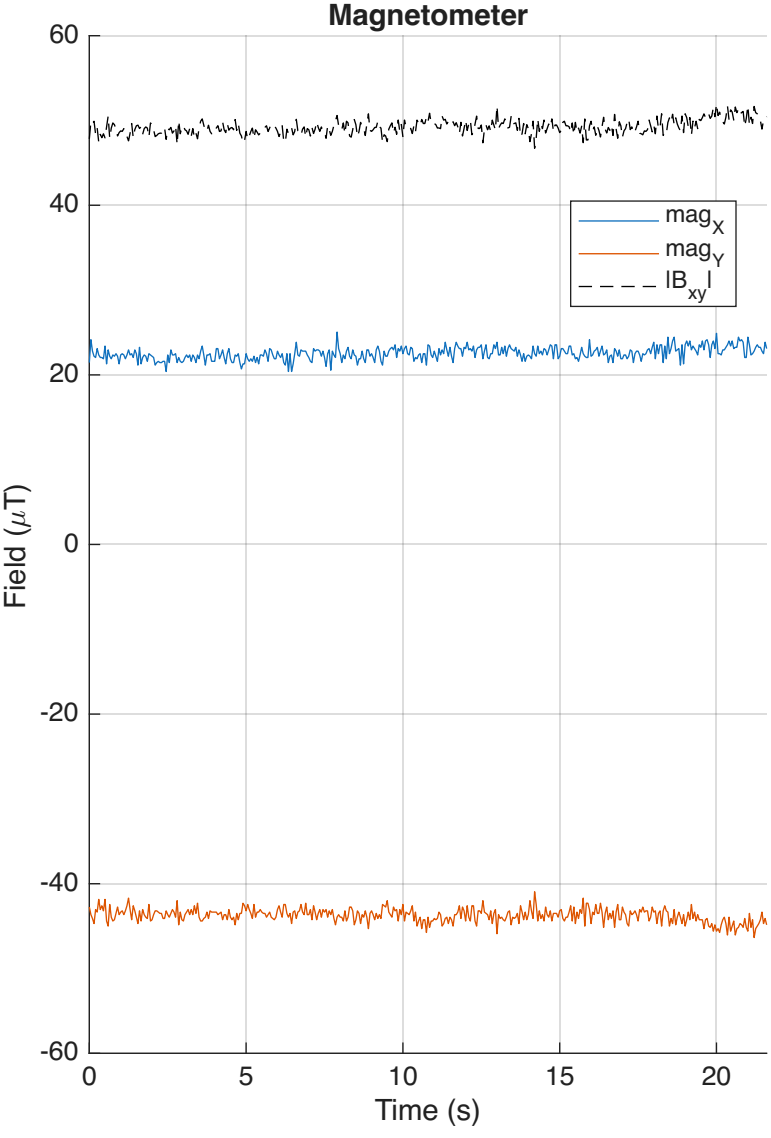
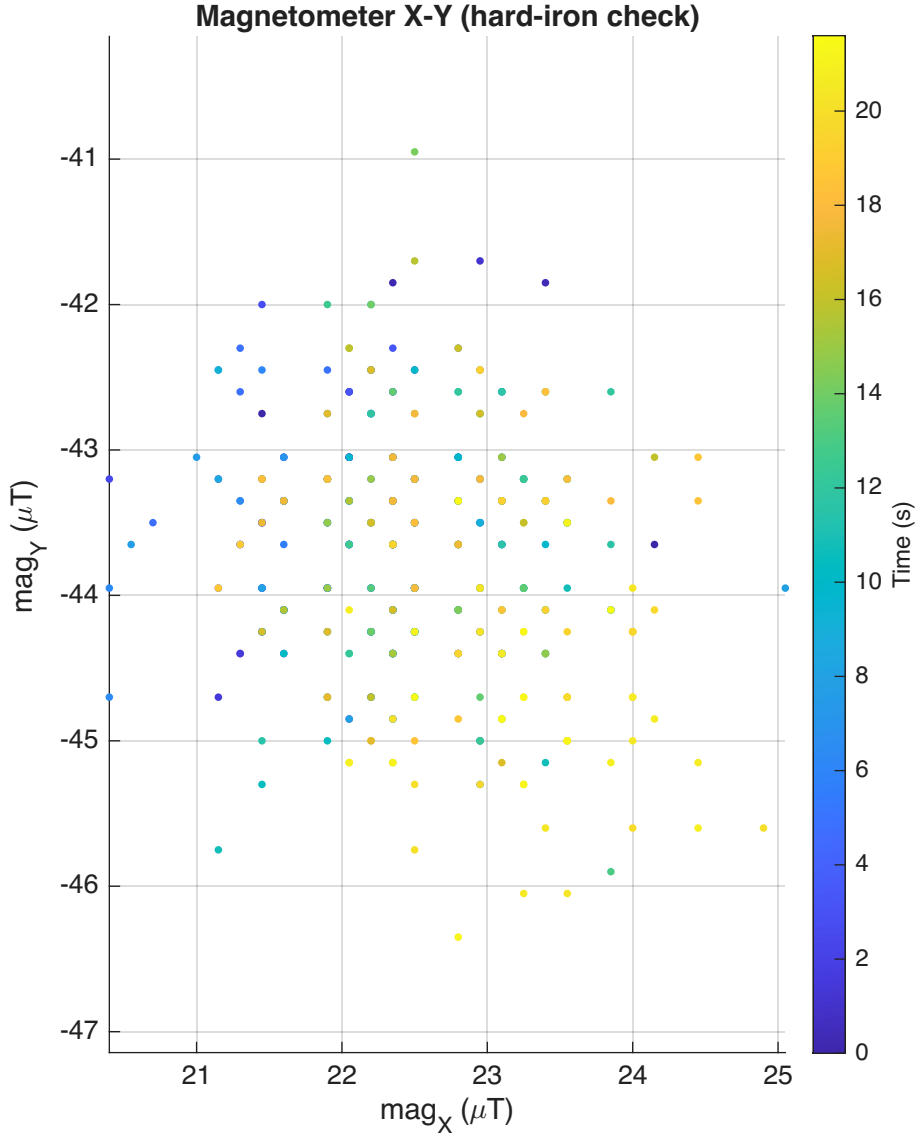

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
2
4
6
8
10
12
14
16
18
20
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
Time (s)

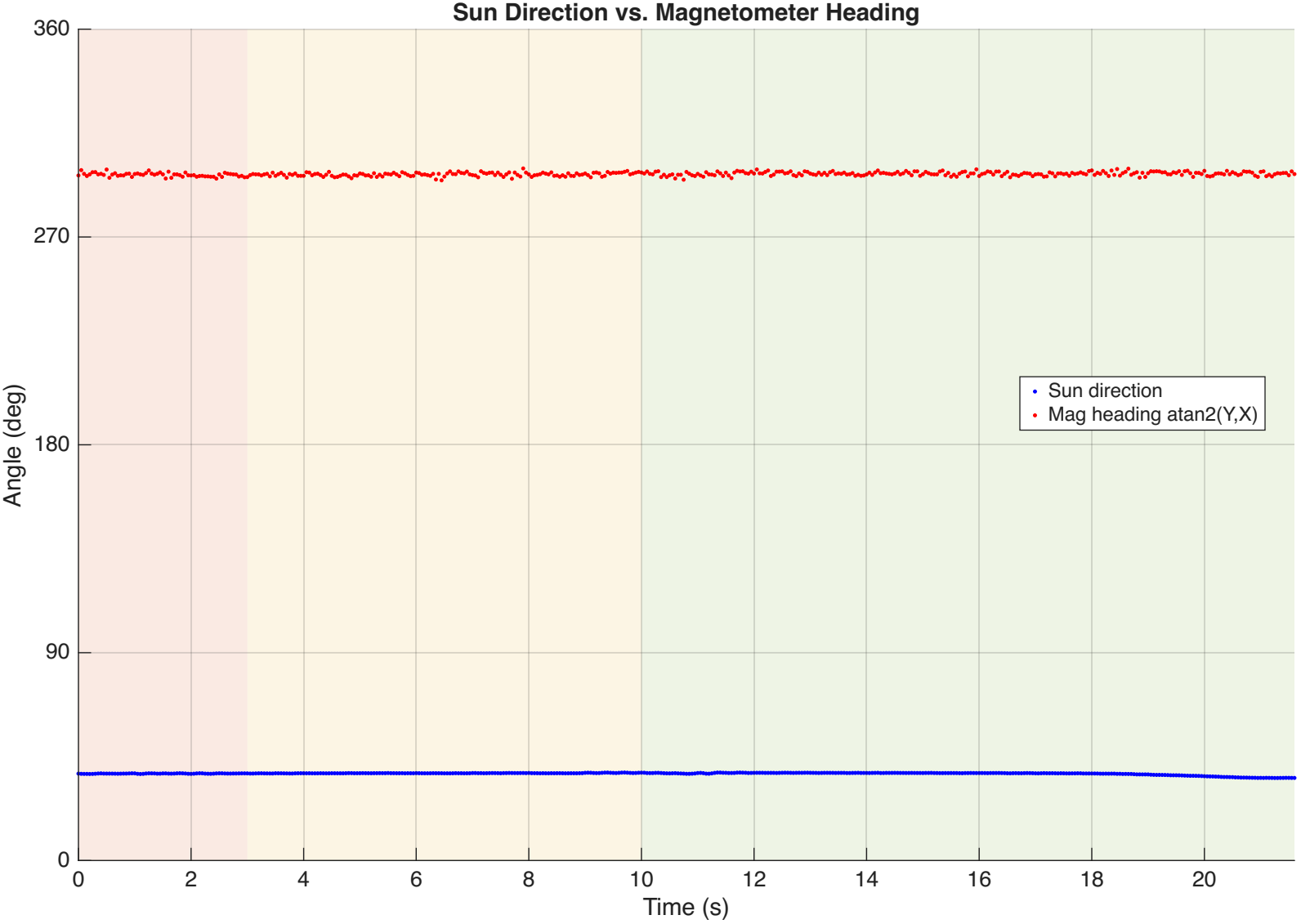

Z-Axis Body Rate (Gyro)
SPINUP
HOLD
TRACK
Body rate (deg/s)
Time (s)
8
6
4
2
0
-2
-4
-6
-8
0
2
4
6
8
10
12
14
16
18
20

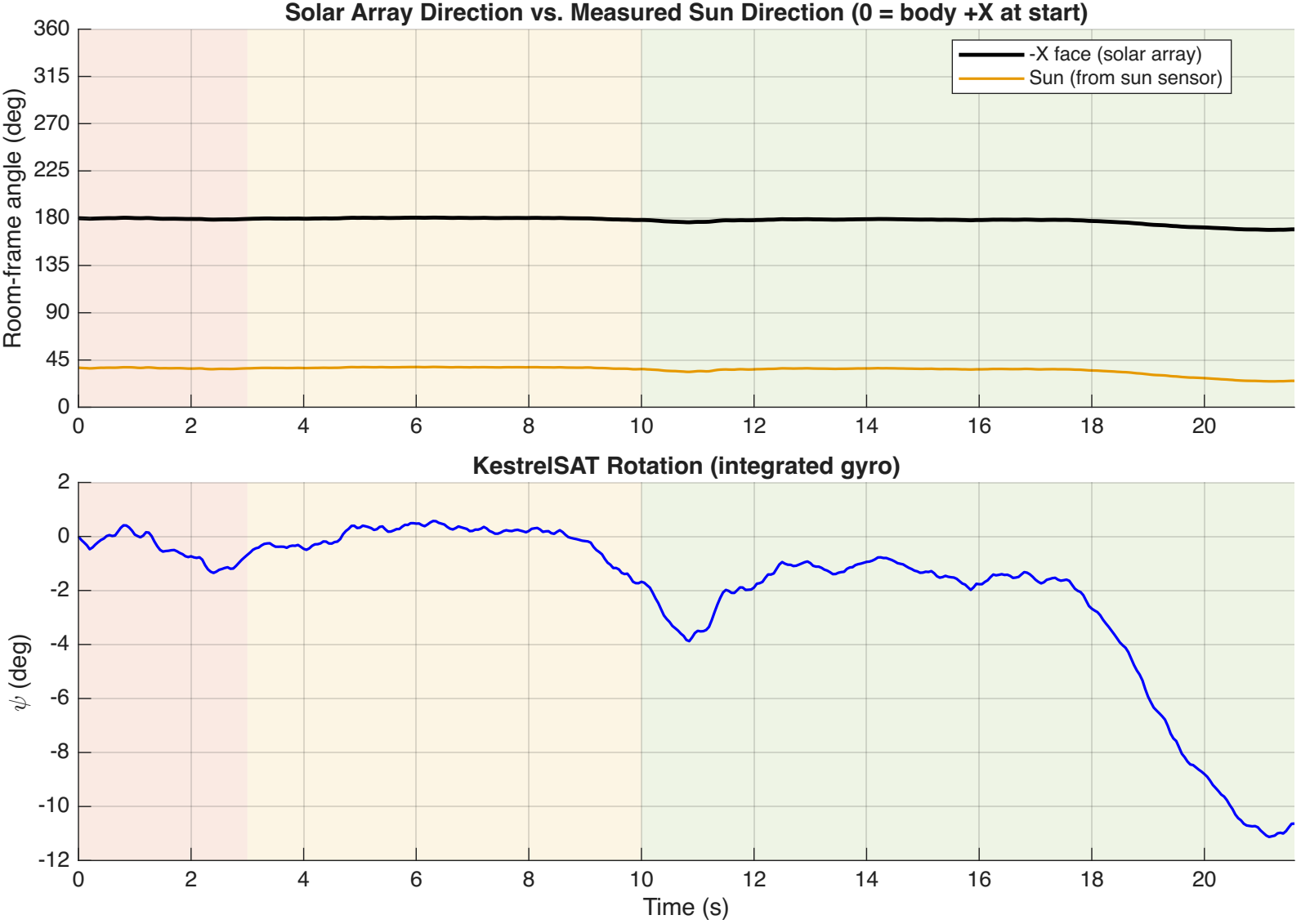

Reaction Wheel Speed
Commanded
Measured
Wheel speed (RPM)
400
350
300
250
200
150
100
50
0
0
2
4
6
8
10
12
14
16
18
20
Commanded Torque (magenta = wheel saturated)
Torque cmd
Saturated
Torque cmd (mN·m)
0
-2
-4
-6
-8
-10
Time (s)

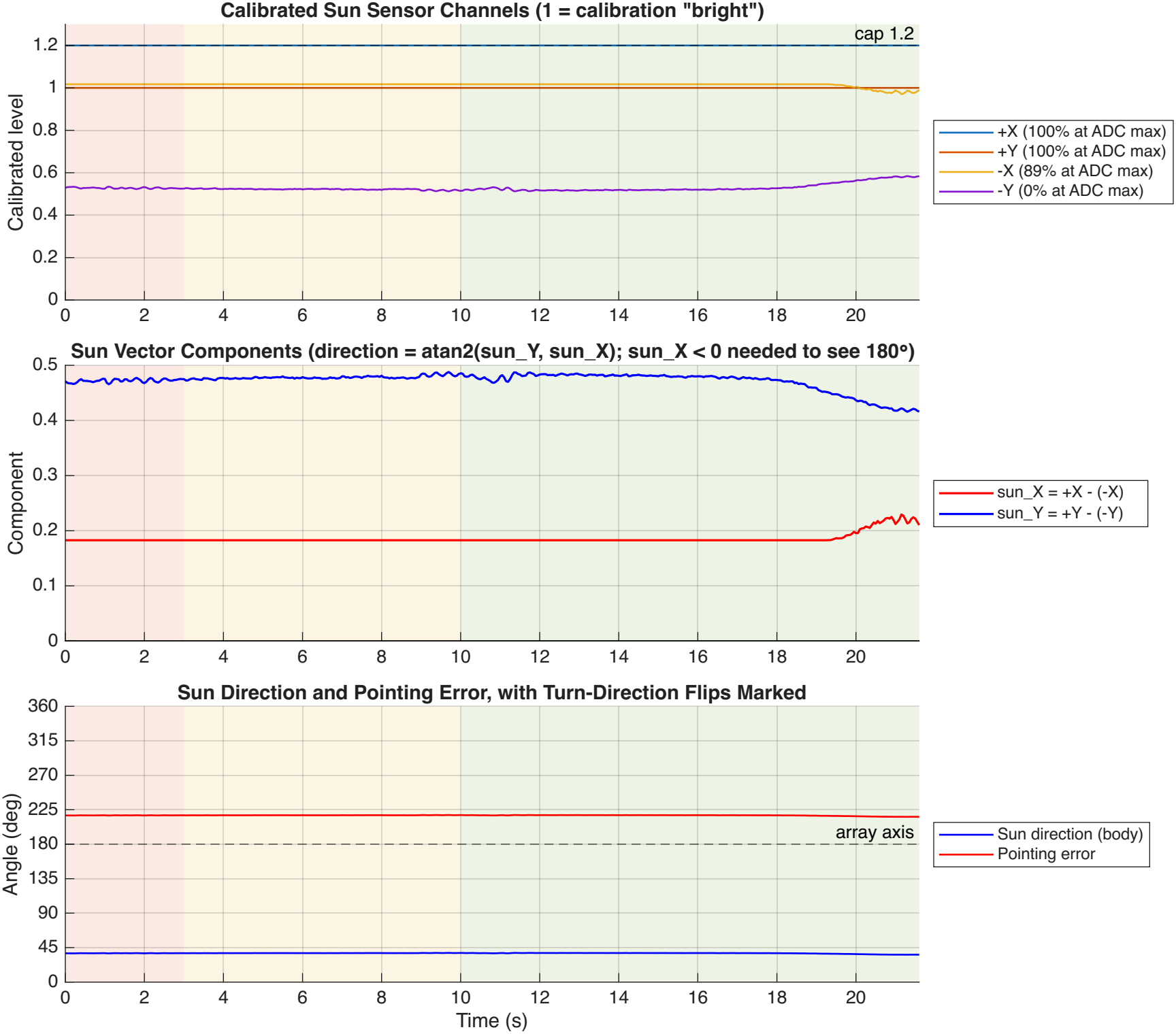

Sun Sensor Photodiode Readings
ADC counts
4500
4000
3500
3000
2500
2000
ADC max
+X
+Y
-X
-Y
0
2
4
6
8
10
12
14
16
18
20
Sun Vector Magnitude
sun_mag
9.12
9.1
9.08
9.06
9.04
9.02
9
8.98
8.96
Time (s)

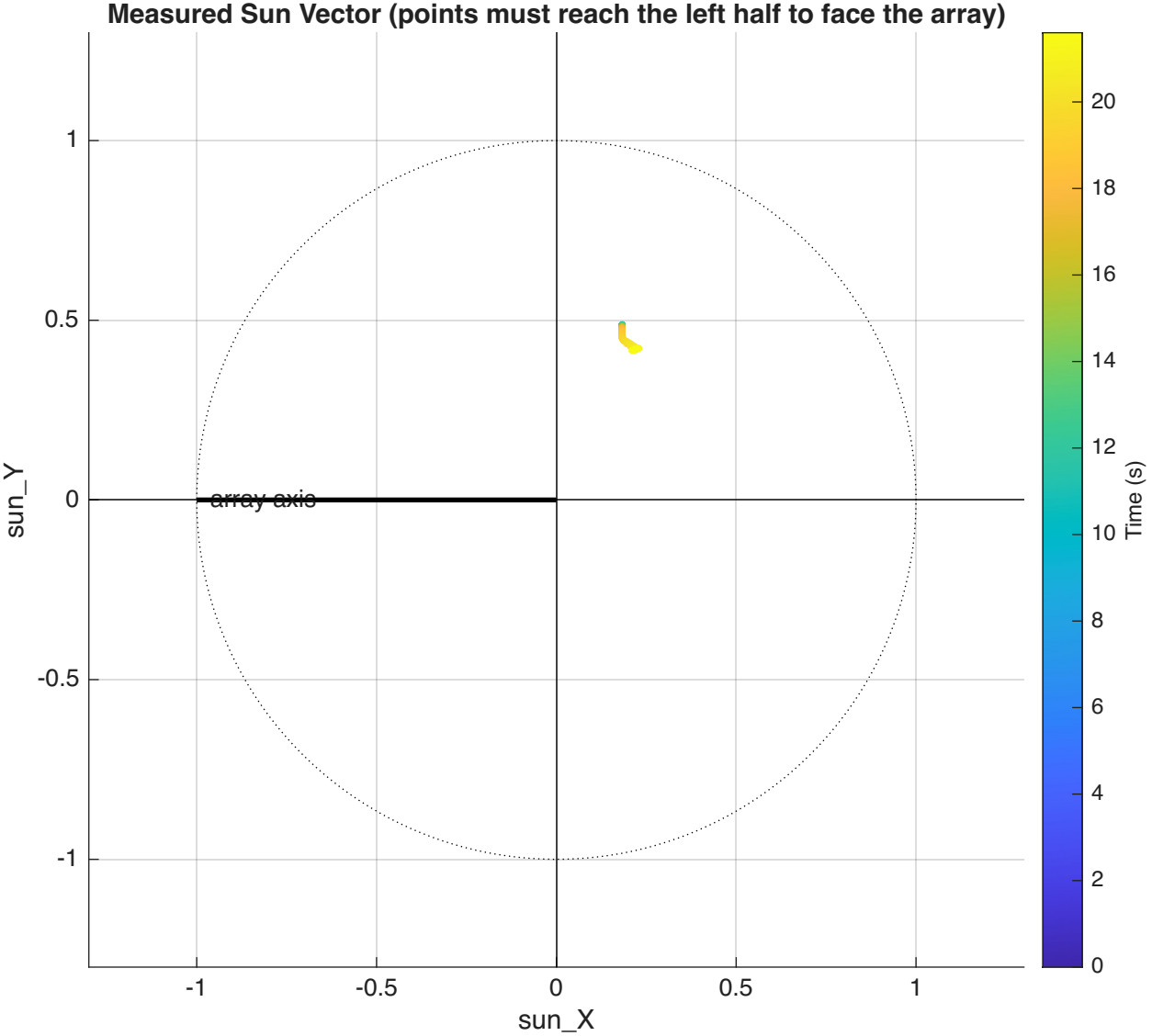

Magnetometer
60
40
20
0
-20
-40
-60
Field (μT)
0
5
10
15
20
Time (s)
mag$_X$
mag$_Y$
|B$_{xy}$|

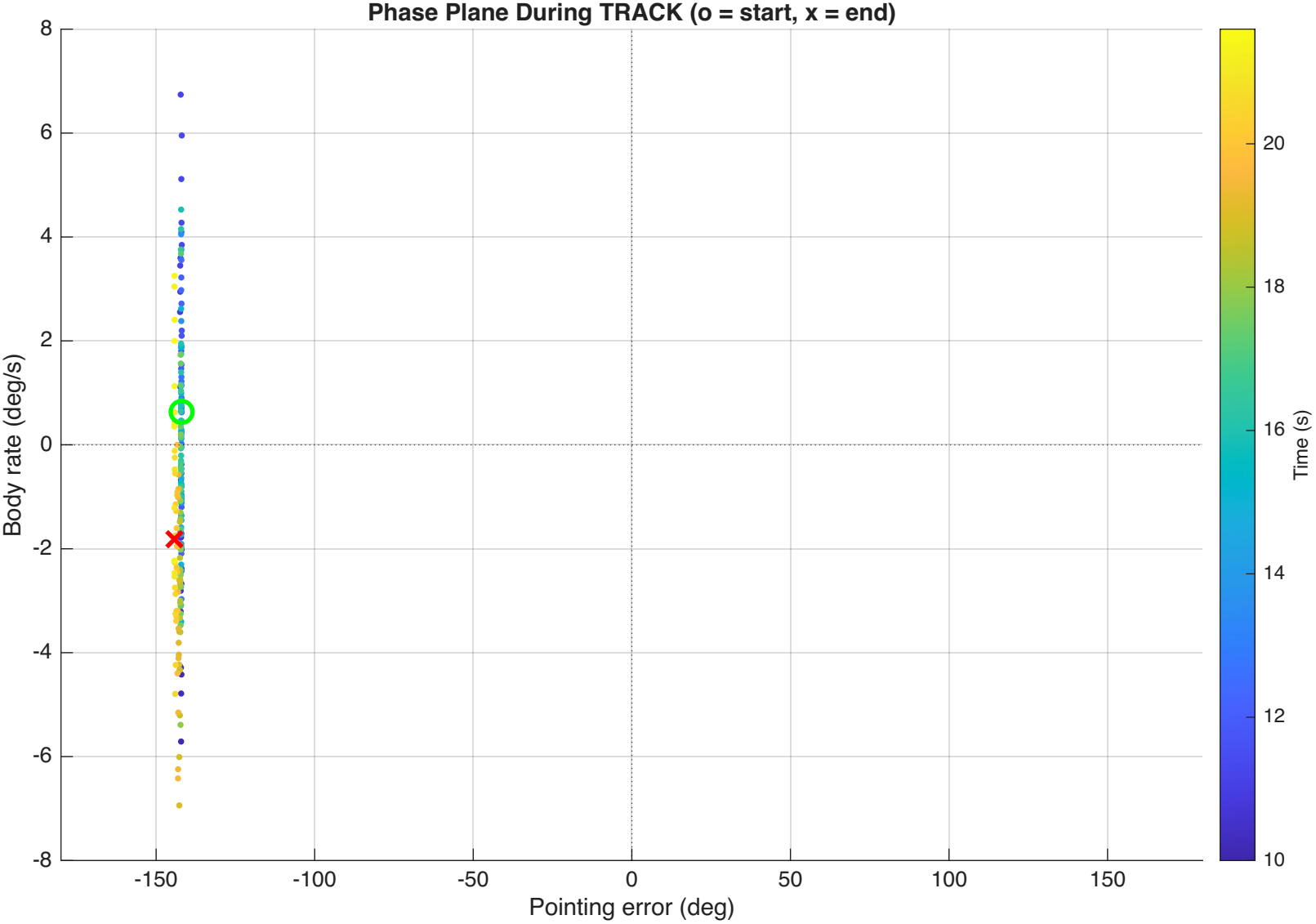

Magnetometer X-Y (hard-iron check)
-41
-42
-43
-44
-45
-46
-47
mag$_Y$ (μT)
21
22
23
24
25
mag$_X$ (μT)
20
18
16
14
12
10
8
6
4
2
0
Time (s)

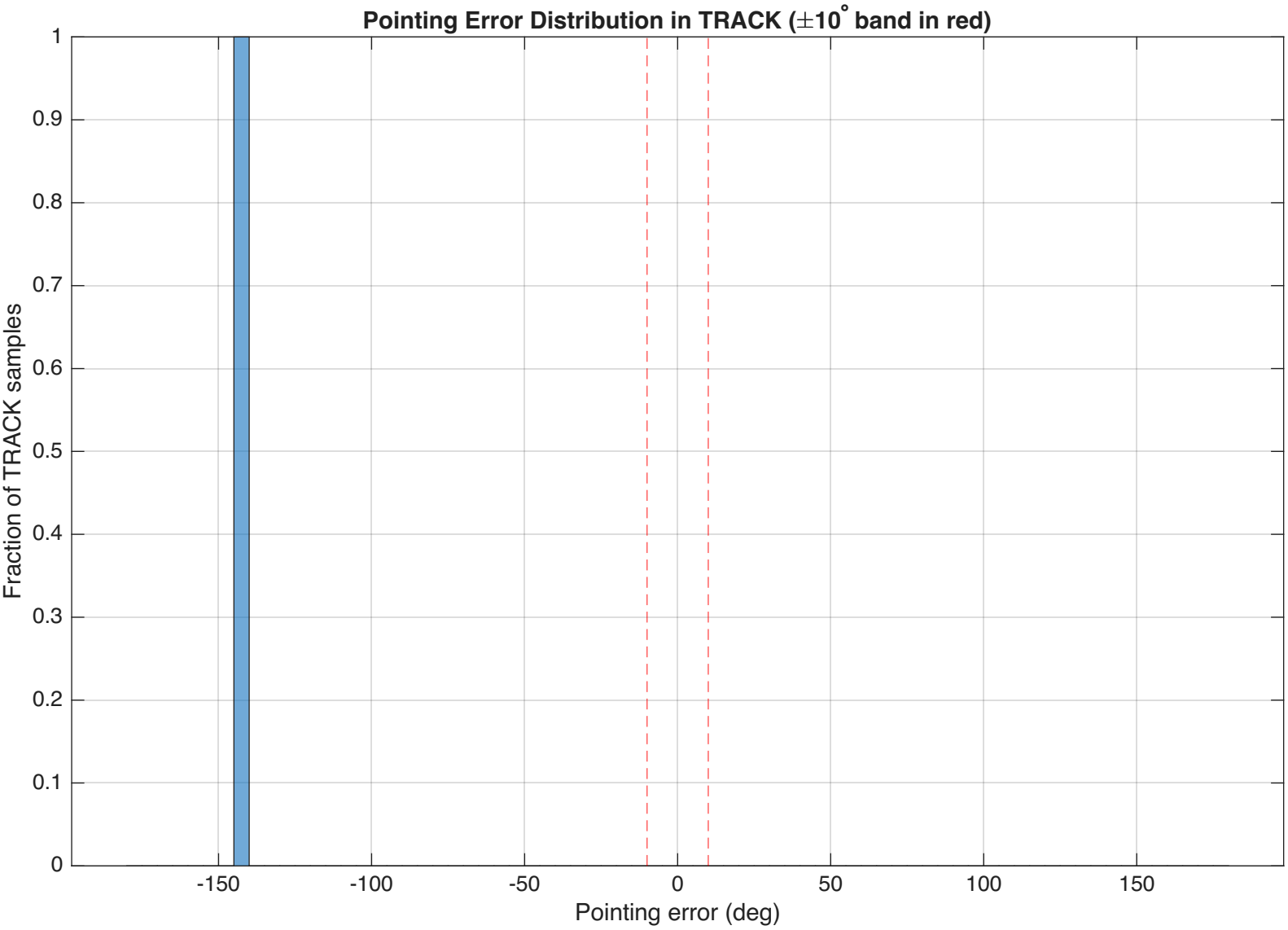

Sun Direction vs. Magnetometer Heading
360
270
180
90
0
Angle (deg)
0
2
4
6
8
10
12
14
16
18
20
Time (s)
Sun direction
Mag heading atan2(Y,X)


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
-X face (solar array)
Sun (from sun sensor)
Room-frame angle (deg)
360
315
270
225
180
135
90
45
0
0
2
4
6
8
10
12
14
16
18
20
KestrelSAT Rotation (integrated gyro)
ψ (deg)
2
0
-2
-4
-6
-8
-10
-12
Time (s)


Calibrated Sun Sensor Channels (1 = calibration "bright")
cap 1.2
Calibrated level
+X (100% at ADC max)
+Y (100% at ADC max)
-X (89% at ADC max)
-Y (0% at ADC max)
Sun Vector Components (direction = atan2(sun_Y, sun_X); sun_X < 0 needed to see 180°)
Component
sun_X = +X - (-X)
sun_Y = +Y - (-Y)
Sun Direction and Pointing Error, with Turn-Direction Flips Marked
Angle (deg)
array axis
Sun direction (body)
Pointing error
Time (s)


Measured Sun Vector (points must reach the left half to face the array)
array axis
sun_Y
sun_X
Time (s)
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
20
18
16
14
12
10
8
6
4
2
0


Phase Plane During TRACK (o = start, x = end)
Body rate (deg/s)
Pointing error (deg)
Time (s)
8
6
4
2
0
-2
-4
-6
-8
-150
-100
-50
0
50
100
150
20
18
16
14
12
10


Pointing Error Distribution in TRACK (±10° band in red)
Fraction of TRACK samples
Pointing error (deg)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-150
-100
-50
0
50
100
150